Simulation Arena

# Deal Aging & Multi-Quarter Sales Cycles: The Pipeline Hopper

**Sales says quarter 2; the odds from your history are 0.00041% for closing then - 0.00061% of the value slips past that date, and >99.9% never closes at all. Plan on $32 of this $3.00M deal, not the full amount. That is after allowing for this deal having sat 5 quarters - a deal that had just arrived would be at 0.0029%.**

Seed 42 | Run run_8807745da7b4 | 100,958 scenarios tested | Markov chain on your own stage-to-stage… | 22 Sep 2026, 07:40

Same seed, same numbers - this run can be reproduced exactly.

## How this works

Every deal on your board hops from stage to stage. We counted how your own deals actually hopped - forward, staying put for a quarter, or dying - and turned those counts into odds. We also account for how long a deal has been sitting still: the older it gets, the more likely it is to slip or die, so a deal that has been parked for a year is priced differently from one that has just arrived. Then we rolled the odds forward one quarter at a time, which gives the chance of the deal being anywhere on the board in each future quarter. Money is then weighted by those odds: if there is a 1-in-8 chance of closing next quarter, only an eighth of the value deserves to be in next quarter's plan.

## The numbers that matter

**CHANCE IT CLOSES IN Q2, AS SALES PROMISED**
**0.00041%**
A slim chance the money actually lands in the quarter it was promised for.
vs if we ignored how long it has been sitting: 0.0029%

**CHANCE IT SLIPS PAST THAT QUARTER**
**0.00061%**
The chance the deal does close, but later than the date sales promised.

**CHANCE IT DIES INSTEAD**
**>99.9%**
The deal never closes - it stalls or the customer walks.
vs 0.0023% still stuck in Legal at the end of the horizon

**WHAT YOU CAN BANK ON BY Q2**
**$32**
The deal value weighted by its real chance of arriving - 0.0% of $3.00M.

**CASH COMMITTED AGAINST ODDS**
**$3.00M**
You are planning to spend $3.00M that has only a 0.0% chance of arriving.
vs $3.00M promised vs $32 realistic

**QUARTERS UNTIL IT REALLY CLOSES**
**3.5**
On average, counted from today, for the deals that do close (3.2 quarters to a final yes or no).
vs sales promised 2

**HOW LONG THIS DEAL HAS BEEN SITTING**
**5 quarters**
The older a deal gets, the more likely it is to slip or die - we have priced that in.
vs we treat it as stale after 4 quarters

## How we got there

### 1. What we measured

We read 100,958 real stage moves from your history and turned them into odds: from each stage a deal moves forward, sits still for a quarter, or dies.

Evidence: 100,958 observed moves

### 2. The forecast versus the odds

Sales has this deal closing in Q2. The odds from your history put it at 0.0% for landing in that quarter - and a >99.9% chance of never landing at all.

Evidence: 0.0% vs 100%

**3. Where the time actually goes**

Deals do not walk through the stages, they sit in them. Looking across the whole horizon, 0.0% of the deals that close do so later than Q2, and 0.0% close after this fiscal year has ended. Even at the end of Q6, 0.0% are still sitting in Legal.

Evidence: 0.0% stuck in Legal

**4. Why the age of the deal matters**

This deal has been running 5 quarters, and we priced it that way. A deal that had only just arrived would show 0.0% for Q2 instead of 0.0%. Deals that sit still are the ones that quietly die, which is why age is part of the forecast rather than an afterthought.

Evidence: 0.0% at 5 quarters vs 0.0% when fresh

**5. The number to plan with**

Weighted by the odds, this deal is worth $32 to Q2's plan, not $3.00M. The difference, $3.00M, is money you would be committing before it has any real chance of arriving.

Evidence: $3.00M of over-commitment

**6. What to do about it**

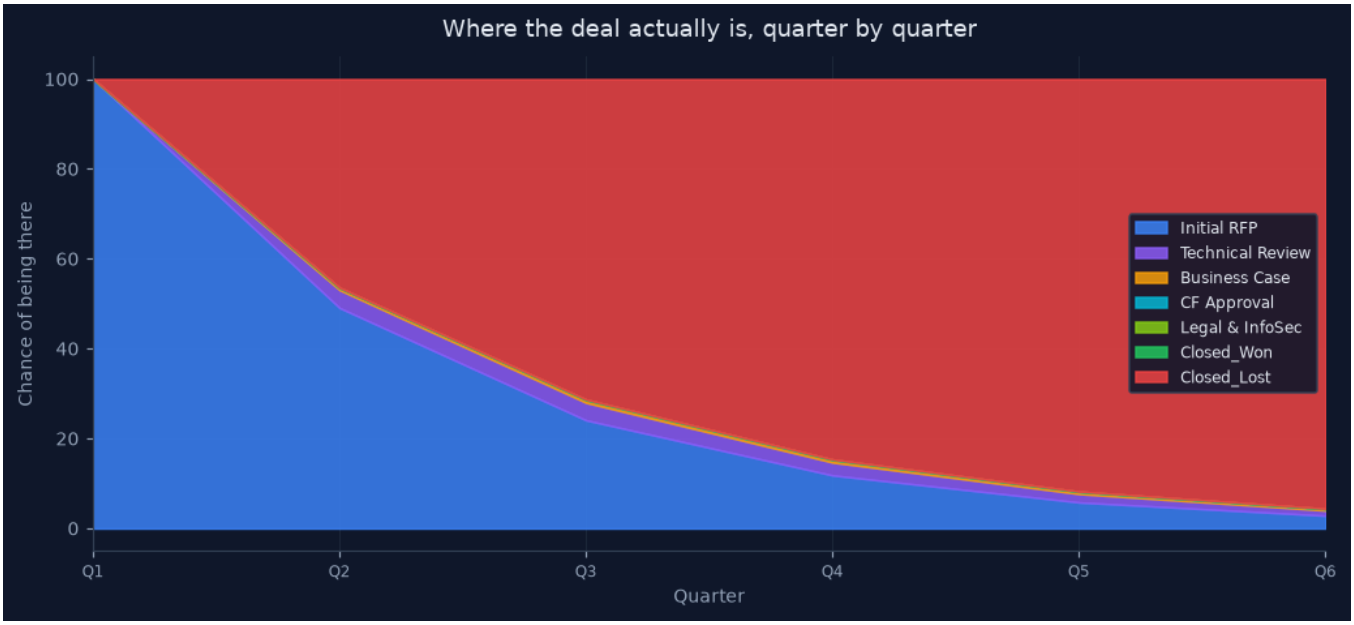
Plan the cash against $32, hold the rest as a contingency, and put the deals that have been sitting the longest in front of whoever owns the stage they are stuck in. A forecast that says 'Q2' with no odds attached is not a forecast - it is a hope.

Evidence: plan on 0.0% of the value

## Charts

### Where the deal actually is, quarter by quarter

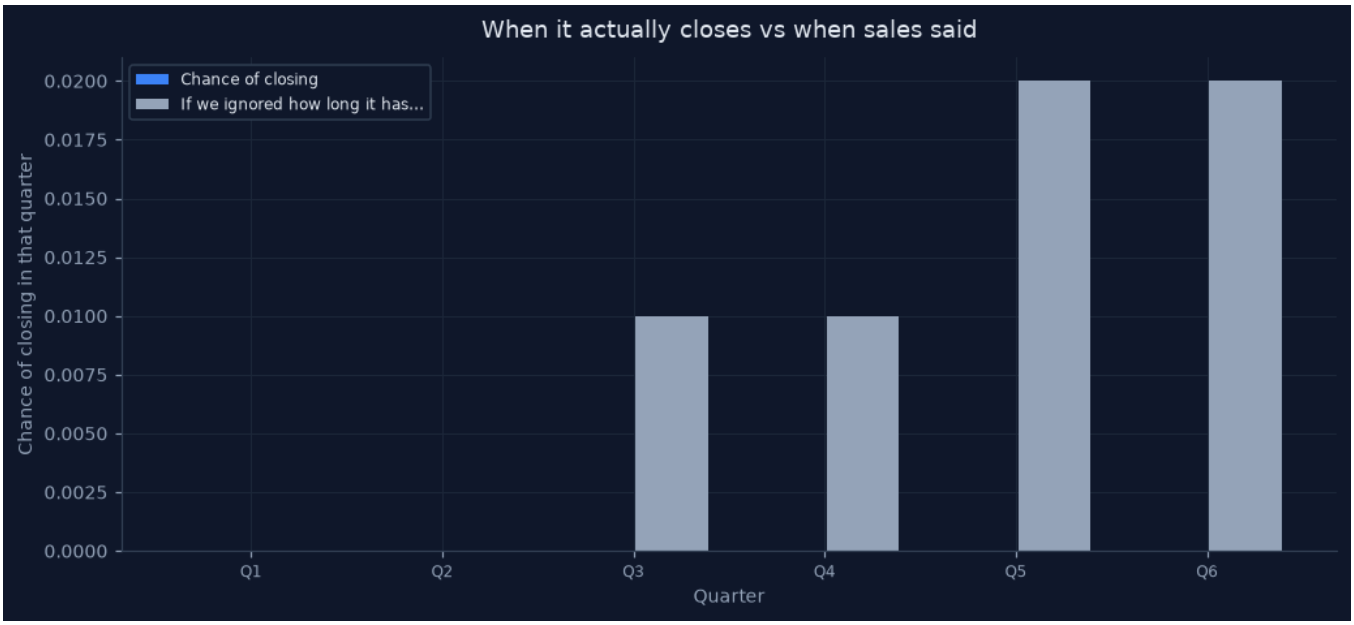
Starting from 'Initial RFP'. Green is won, red is dead.




*What this shows: Read the green band: that is the chance the deal is already closed by each quarter. The grey blocks are the stages it is still sitting in.*

### When it actually closes vs when sales said

Chance of closing in each quarter; sales promised Q2




*What this shows: The tall bar is the quarter the deal is most likely to close in. It is almost never the quarter on the forecast.*

### Money you can actually plan on

Cumulative deal value weighted by its chance of arriving




*What this shows: The gap between the green line and the red line is cash you would be committing before it has arrived.*

## Tables

### Quarter by quarter: where the deal is

The chance of the deal sitting in each stage at the end of every quarter.

| Quarter | Initial RFP | Technical Review | Business Case | CF Approval | Legal & InfoSec | Closed_Won | Closed_Lost |
|---|---|---|---|---|---|---|---|
| Q1 | 49.1% | 4.0% | 0.48% | 0.055% | 0.00066% | 0.00066% | 46.4% |
| Q2 | 24.1% | 3.9% | 0.63% | 0.093% | 0.003% | 0.0011% | 71.3% |
| Q3 | 11.8% | 2.9% | 0.57% | 0.1% | 0.004% | 0.0013% | 84.6% |
| Q4 | 5.8% | 1.9% | 0.45% | 0.091% | 0.0038% | 0.0015% | 91.8% |
| Q5 | 2.8% | 1.1% | 0.32% | 0.072% | 0.0031% | 0.0016% | 95.6% |
| Q6 | 1.4% | 0.67% | 0.21% | 0.053% | 0.0023% | 0.0016% | 97.7% |

Read the 'Closed_Won' column to see how the chance of closing builds up.

### The money, weighted by the odds

What this deal is really worth to each quarter's plan.

| Quarter | Chance of closing then | Value earned that quarter | Value banked by then | Chance closed by then | Chance dead by then | Still not banked |
|---|---|---|---|---|---|---|
| Q1 | 0.00066% | $20 | $20 | 0.00066% | 46.4% | $3.00M |
| Q2 | 0.00041% | $12 | $32 | 0.0011% | 71.3% | $3.00M |
| Q3 | 0.00025% | $8 | $40 | 0.0013% | 84.6% | $3.00M |
| Q4 | 0.00015% | $5 | $44 | 0.0015% | 91.8% | $3.00M |
| Q5 | <0.0001% | $3 | $47 | 0.0016% | 95.6% | $3.00M |
| Q6 | <0.0001% | $2 | $48 | 0.0016% | 97.7% | $3.00M |

Row 2 is the quarter sales promised.

### What sitting still costs you

The same deal judged at three ages, plus the age we actually used for it.

| How long it has been sitting | Chance it closes by Q2 | Chance it never closes |
|---|---|---|
| Fresh (under 2 quarters) | 0.025% | 98.1% |
| Settled (2-4 quarters) | 0.01% | >99.9% |
| Stale (over 4 quarters) | <0.0001% | 100.0% |
| This deal (5 quarters) | 0.0011% | >99.9% |

The older a deal is, the lower it sits. The highlighted row is the age we used for the forecast above.

### Your open deals, weighted the same way (top 25)

Every open deal on the board, priced by the odds of it actually closing - each one judged at its own age.

| Deal | Stage now | Value | Closes next quarter | Within 2 quarters | Ever closes | Expected quarter | Value you can bank on | Quarters sitting |
|---|---|---|---|---|---|---|---|---|
| P-00011589 | Business Case | $40.00M | 0.08% | 0.29% | 3.5% | 6.7 | $1.40M | 1 |
| P-00038587 | Initial RFP | $40.00M | 0.0071% | 0.025% | 1.9% | 9.5 | $762k | 0 |
| P-00054552 | Business Case | $40.00M | <0.0001% | <0.0001% | <0.0001% | n/a | $0 | 7 |
| P-00047903 | Initial RFP | $40.00M | 0.0071% | 0.025% | 1.9% | 9.5 | $762k | 0 |
| P-00054879 | Technical Review | $40.00M | 0.0042% | 0.053% | 2.3% | 8.3 | $932k | 1 |
| P-00042281 | Technical Review | $36.62M | 0.0042% | 0.053% | 2.3% | 8.3 | $853k | 1 |
| P-00027130 | Business Case | $35.98M | 0.08% | 0.29% | 3.5% | 6.7 | $1.26M | 1 |
| P-00051237 | Initial RFP | $32.70M | 0.0052% | 0.01% | 0.042% | 5.2 | $14k | 2 |
| P-00068394 | Initial RFP | $32.42M | 0.0071% | 0.025% | 1.9% | 9.5 | $618k | 0 |
| P-00081129 | Technical Review | $29.82M | 0.0083% | 0.017% | 0.062% | 4.5 | $19k | 3 |
| P-00032218 | Initial RFP | $29.58M | 0.0071% | 0.025% | 1.9% | 9.5 | $564k | 1 |
| P-00004141 | Initial RFP | $28.96M | 0.0052% | 0.01% | 0.042% | 5.2 | $12k | 3 |
| P-00026774 | CF Approval | $28.48M | <0.0001% | <0.0001% | <0.0001% | n/a | $0 | 9 |
| P-00049379 | Technical Review | $27.33M | 0.0083% | 0.017% | 0.062% | 4.5 | $17k | 2 |
| P-00059817 | Initial RFP | $26.98M | <0.0001% | <0.0001% | <0.0001% | n/a | $0 | 11 |
| P-00083844 | Initial RFP | $26.24M | 0.0052% | 0.01% | 0.042% | 5.2 | $11k | 2 |
| P-00065601 | Technical Review | $25.72M | 0.0083% | 0.017% | 0.062% | 4.5 | $16k | 2 |

| Deal | Stage now | Value | Closes next quarter | Within 2 quarters | Ever closes | Expected quarter | Value you can bank on | Quarters sitting |
|---|---|---|---|---|---|---|---|---|
| P-00072655 | Business Case | $25.04M | 0.08% | 0.29% | 3.5% | 6.7 | $874k | 1 |
| P-00095862 | Initial RFP | $24.60M | 0.0071% | 0.025% | 1.9% | 9.5 | $469k | 0 |
| P-00085009 | Technical Review | $24.57M | 0.0042% | 0.053% | 2.3% | 8.3 | $572k | 1 |
| P-00075948 | Technical Review | $24.35M | 0.0042% | 0.053% | 2.3% | 8.3 | $567k | 0 |
| P-00004222 | Legal & InfoSec | $24.34M | 0.85% | 1.2% | 1.5% | 1.7 | $362k | 3 |
| P-00044677 | Business Case | $24.14M | 0.08% | 0.29% | 3.5% | 6.7 | $842k | 1 |
| P-00072186 | Technical Review | $24.07M | 0.0083% | 0.017% | 0.062% | 4.5 | $15k | 2 |
| P-00080629 | Initial RFP | $24.05M | 0.0071% | 0.025% | 1.9% | 9.5 | $458k | 0 |

Value you can bank on = deal value x its chance of ever closing.

## What we recommend

**Do not book $3.00M for Q2. Book $32 and hold the rest back.**

**Risk High** 100 out of 100 | Confidence: High

### Why

• The chance of closing in Q2 is 0.0%, not 100%.

• 0.0% of the value arrives later than promised. and >99.9% never arrives at all.

• Planning on the full $3.00M commits $3.00M of cash against a 0.0% chance.

• The deal has been running 5 quarters; at that age a deal is judged 0.0% likely to close by Q2 if it were fresh, so sitting still is costing real odds.

### What to do

**1.** Put $32 in the Q2 plan and treat the remaining $3.00M as contingent on the deal actually reaching the next stage.

**2.** Attach a stage gate to any hiring, inventory or spend that depends on this deal.

**3.** Re-forecast the whole open pipeline the same way, not just this deal - the same optimism is almost certainly in the rest of the board.

**4.** Ask sales to forecast with odds attached, not dates.

**The trade-off:** Booking less now means a smaller number in this quarter's plan - but it avoids the mid-year freeze that follows a promise that does not arrive.

## What we assumed and what to watch

**Worth knowing before you act**

• Sales says Q2; the model says 0.0%. Committing the full $3.00M on that date is the single biggest risk in this plan.

• >99.9% of deals at this stage never close at all - they are not 'slipping', they are gone. Treat that part of the pipeline as zero.

• This deal has been sitting 5 quarters - past the 4-quarter mark where we start treating a deal as stale. Its chance of closing is 0.0% rather than 0.0% for a fresh deal.

• Your file holds 376,539 rows. We are running on an evenly spread sample of 188,272 of them (50.0%), keeping rows intact rather than splitting a record down the middle, because a working set that big does not fit in memory while keeping the tool responsive. Every number below comes from those 188,272 rows.

**Assumptions**

• Learned how deals hop between stages from 100,958 real stage moves (the quarter column told us what happened first). Paths nobody has taken yet are given a small benefit of the doubt rather than a flat zero chance.

• A deal that is won or dead stays that way - we force those stages to be final even if the history is messy.

• We priced how long deals sit still: past 4 quarters a deal gets progressively more likely to slip or die, up to a cap of 90.0% extra risk. The deal being forecast is treated as 5 quarters old.

• Deal value $3,000,000, starting in 'Initial RFP', 6 quarters ahead, fiscal year = 4 quarters.

• Each quarter a deal moves at most one stage forward; it may sit still or die. Deal size does not change the odds.

• Stage behaviour is assumed stable: the pattern in your history continues.

• Columns of your file we actually used: deal_id, stage, stage_order, quarter, amount, entered_quarter, days_in_stage.

• Deal / opportunity ID -> your 'deal_id' column

• Pipeline stage -> your 'stage' column

• Stage order (1,2,3...) -> your 'stage_order' column

• Quarter of this sighting -> your 'quarter' column

• Deal value -> your 'amount' column

• Quarter the deal entered -> your 'entered_quarter' column

• How long the deal has been sitting (quarters) -> not in your file; not used

• How long it has been sitting (days) -> your 'days_in_previous_stage' column

• Data used: 188,272 of the 376,539 rows in your file (50.0%), drawn evenly across it.

## Appendix: the last steps of the run

Every line was produced while the model ran. Re-running with the same seed replays it exactly.

| Time | Stage | What happened |
|---|---|---|
| 0.02s | prepare | You chose 'deal_id' for 'Deal / opportunity ID'. |
| 0.02s | prepare | You chose 'stage' for 'Pipeline stage'. |
| 0.02s | prepare | You chose 'stage_order' for 'Stage order (1,2,3...)'. |
| 0.02s | prepare | You chose 'quarter' for 'Quarter of this sighting'. |
| 0.02s | prepare | You chose 'amount' for 'Deal value'. |
| 0.02s | prepare | You chose 'entered_quarter' for 'Quarter the deal entered'. |
| 0.02s | prepare | We could not find a 'How long the deal has been sitting (quarters)' column. We left it out and will work without it. |
| 0.02s | prepare | You chose 'days_in_previous_stage' for 'How long it has been sitting (days)'. |
| 0.02s | prepare | Deal / opportunity ID -> your 'deal_id' column |
| 0.02s | prepare | Pipeline stage -> your 'stage' column |
| 0.02s | prepare | Stage order (1,2,3...) -> your 'stage_order' column |
| 0.02s | prepare | Quarter of this sighting -> your 'quarter' column |
| 0.02s | prepare | Deal value -> your 'amount' column |
| 0.02s | prepare | Quarter the deal entered -> your 'entered_quarter' column |
| 0.02s | prepare | How long the deal has been sitting (quarters) -> not in your file; not used |
| 0.02s | prepare | How long it has been sitting (days) -> your 'days_in_previous_stage' column |

| Time | Stage | What happened |
|---|---|---|
| 3.17s | prepare | Read 188,272 records into the model. |
| 3.17s | simulate | Running the simulation… |
| 3.17s | read | Reading how your deals have moved through the pipeline... |
| 5.01s | prepare | Laid the stages out in the order you gave us. |
| 5.01s | prepare | Learned the hop pattern from 100,958 real stage moves across 87,314 deals. |
| 5.01s | prepare | Deals that sit still get riskier: this one has been going 5 quarters, so we priced it against the stale-deal pattern. |
| 5.01s | simulate | Rolling the odds forward 6 quarters from 'Initial RFP'... |
| 5.01s | summarize | Sales says quarter 2. The odds from your history are 0.0% for closing in that quarter. |
| 6.05s | summarize | Done. Plan on $32 of $3.00M for Q2, not the full amount. |